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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Falcons
Selected Domain	Domain 2 – Orbyte Smart SME Challenge
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth
Project / Solution Name	KiraOrder for Orbyte
Name (Leader)	ZHOU JINGXI
Email & Phone (Leader)	DSC2509065@xmu.edu.my | +60 174842902
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Malaysian wholesale SMEs receive WhatsApp orders mixing Malay, English and Chinese, abbreviations and customer-specific product names. Clerks must identify products, interpret quantities and packs, check prices and stock, clarify requests and prepare delivery orders. Ambiguity creates rework, stock disputes and avoidable follow-up.

KiraOrder addresses the gap between an incoming message and an approved order. Clerks need less transcription; owners need control over exceptions; warehouse staff need accurate picking information without unnecessary financial or contact details.

The problem is grounded in the official SDC Domain 2 Orbyte Smart SME Challenge brief (pages 3–4) and team domain research. We have not completed independent SME interviews or a field pilot. Our planning baseline is approximately 210 seconds median handling time per order, with up to 30% transcription or stockout discrepancies. These are team estimates, not measured findings. Both assumptions require pilot validation. The primary intended impact is economic and operational: improved SME productivity and reliable order records, aligned with SDG 8.2 and 8.3.

2.2 The Proposed AI Solution

KiraOrder turns pasted WhatsApp-style orders into reviewable drafts and approved delivery orders through five stages: Trigger, Connected Data, Decision Logic, Human Approval and Visible Outcome.

A pasted message is the trigger. Tenant catalogues, customer aliases, prices and inventory provide connected data. Exact aliases can resolve directly; the configured AI path uses Qwen2.5-7B-Instruct-AWQ on RunPod to extract product-family candidates, quantities, units, packs and conditional phrases. Backend rules calculate prices and check stock. Data-settled conditions are resolved by rules; unclear requests go back to the customer; price exceptions go to the owner; routine orders remain with the clerk. Human approval produces a delivery order, restricted warehouse view and audit trail.

The model is restricted to supplied family IDs or UNKNOWN, must preserve source spans and cannot set prices or approve orders. Uncertain interpretations remain visible for correction.

React/TypeScript, FastAPI, PostgreSQL and vLLM apply the GenAI API, database and cloud-hosting skills taught in the SDC workshops. The deployment owner reports AI_PROVIDER=runpod_adapter, with RunPod Active=0 and public parse falling back; this is not an integrated live-AI demonstration. Local Demo is a separate browser-only mode using synthetic data. RunPod inference has been exercised independently.

Development evidence records 23 live LLM extraction cases with zero fallback; accuracy is NOT_COMPUTED because the development inputs lack an independent answer key; no accuracy claim is made. The TEST32 freeze manifest was approved and merged in PR #65 on 24 October, but 12 workflow cases remain NOT_RUN. Phase 2 will complete integration and evaluate handling time, clarification quality and first-use completion.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

The React/TypeScript frontend calls FastAPI on Replit. FastAPI uses Neon PostgreSQL for drafts, approvals, delivery orders and audit records. A separate configured extraction branch invokes Qwen2.5-7B-Instruct-AWQ through vLLM on RunPod after masking identifying text. Figure A1 shows these branches.

He Minghao's latest deployment update reports AI_PROVIDER=runpod_adapter, Active=0 and public parse falling back. This is not an integrated live-AI demonstration; the repository's StubExtractor note describes an earlier configuration. Local Demo uses synthetic browser data without backend or model calls. Tenant isolation, role checks, transaction controls and human approval support the workflow.

Official sponsor resources used: US$25 RunPod credit for model hosting and evaluation, and Ye Lianghui's Replit sponsored Agent usage for frontend development. He Minghao's Replit Core deployment is separately self-funded; Neon uses its Free plan.

Primary frontend: https://hozier123.github.io/sdc-2026-team-project/
Legacy frontend / old UI: https://kiraorder-web.replit.app
API: https://kiraorder-api.replit.app/api/v1
Repository: https://github.com/Hozier123/sdc-2026-team-project

3.2 Design-Thinking & Deployment Strategy

We structured the prototype around the clerk preparing orders, the owner deciding exceptions and the warehouse user fulfilling approved orders. Uncertain products, units and conditional requests remain visible before approval. Synthetic multilingual fixtures exercise ambiguity, shortages and price conditions without using real customer records.

The official challenge brief establishes the problem context; it does not substitute for independent SME validation. No completed external SME pilot is claimed. Interface checks and backend tests support development, while Phase 2 will compare manual and assisted handling on equivalent orders and record correction effort, completion time and first-use success.

Deployment separates the frontend, API, database and model service. The RunPod adapter configuration and live model availability must be verified end to end before claiming an integrated live-AI demonstration. Pilot preparation also requires tested access instructions, model warm-up, reviewed workflow preconditions, manual data-cleanup responsibility and a clear explanation of known limitations. TEST32's approved freeze manifest fixes evaluation scope; it does not turn unexecuted workflow cases into passing results.

4. Preliminary Environmental & SDG Impact Framework

Metric Category	Your Response
Target SDG sub-targets	SDG 8.2: productivity through technological upgrading. SDG 8.3: support SME productive activity through more reliable order records. Source: United Nations Goal 8, https://sdgs.un.org/goals/goal8. No material environmental impact is claimed; primary impact is economic/operational.
Current problem baseline	Largely manual WhatsApp order transcription. Team planning estimates (no measurement yet), 24 October 2026: approximately 210 seconds median per order and up to 30% transcription/stockout discrepancies. These are unvalidated assumptions. A pilot must establish timing, sample size and discrepancy definitions before comparison.
Core measurable KPI	Median seconds from reading an order to a correct, approved delivery order, including clarification and correction. Track silent product/quantity mismatches, price-condition recall and first-use completion without guidance. Report denominators separately for AI extraction and end-to-end workflow.
Phase 2 impact target	At least 50% lower median handling time against a pilot-validated manual baseline; zero observed silent mismatches in the evaluated sample; human handling for all detected price conditions; at least 80% first-use completion without guidance. A validated 210-second baseline would imply a 105-second time target. These are targets, not achieved results.

5. Eligibility & Readiness Checklist

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

The team confirms that the problem is supported by the official challenge brief and team research. SME interviews and a field pilot have not been completed; baseline figures remain planning estimates.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

No. No team member's pre-existing work is included.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

Yes. Third-party resources are declared below and detailed in the accompanying Technical and Tools Declaration.

Name	**Type (code/dataset/model)**	**License**
Qwen2.5-7B-Instruct-AWQ; vLLM 0.32.0	Model; serving code	Apache-2.0
React, React DOM, Three.js, React Three Fiber / Drei; Vite and React plugin; Tailwind and Vite plugin	Frontend code	MIT
AJV, ajv-formats, openapi-typescript, tsx; React / DOM / Three type declarations	Frontend tooling	MIT
FastAPI, Pydantic, pydantic-settings, RapidFuzz; pytest, Ruff, jsonschema, PyYAML, Hatchling	Backend and tooling	MIT
TypeScript 5.8.3; official SDC repository starter	Code; scaffold	Apache-2.0
HTTPX 0.28.1; Uvicorn 0.54.0; Starlette 1.9.0	Code	BSD-3-Clause
pglast 7.21	Runtime SQL parser	GPL-3.0-or-later
psycopg 3.3.6	Database driver	LGPL-3.0
Python 3.13; Node.js 22; PostgreSQL 17.11	Runtimes; database	PSF; MIT; PostgreSQL License

Name	Type (code/dataset/model)	License
GSAP 3.15.0; @gsap/react 2.1.2; lucide-react 0.468.0	Animation; icons	GSAP Standard License; ISC for Lucide
React Bits DepthText, TS-CSS; retrieved 23 October 2026	Opening wordmark source component	MIT + Commons Clause License Condition v1.0
Marcellus; Cormorant Garamond; retained Cinzel	Font assets	SIL Open Font License 1.1
Background video, posters, mobile assets and nebula.png	Visual assets; see tools declaration Appendix F	Origin / permission evidence remains incomplete

Full dependency versions, source notices and visual-asset provenance are provided in the accompanying Technical and Tools Declaration. Evaluation and demo datasets are team-created synthetic fixtures. No model fine-tuning or external training dataset is claimed.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

No. The team reports no employment-related ownership conflicts.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Team leader: ZHOU JINGXI
Signature:
Date signed: 2026/10/25

Appendix A System architecture

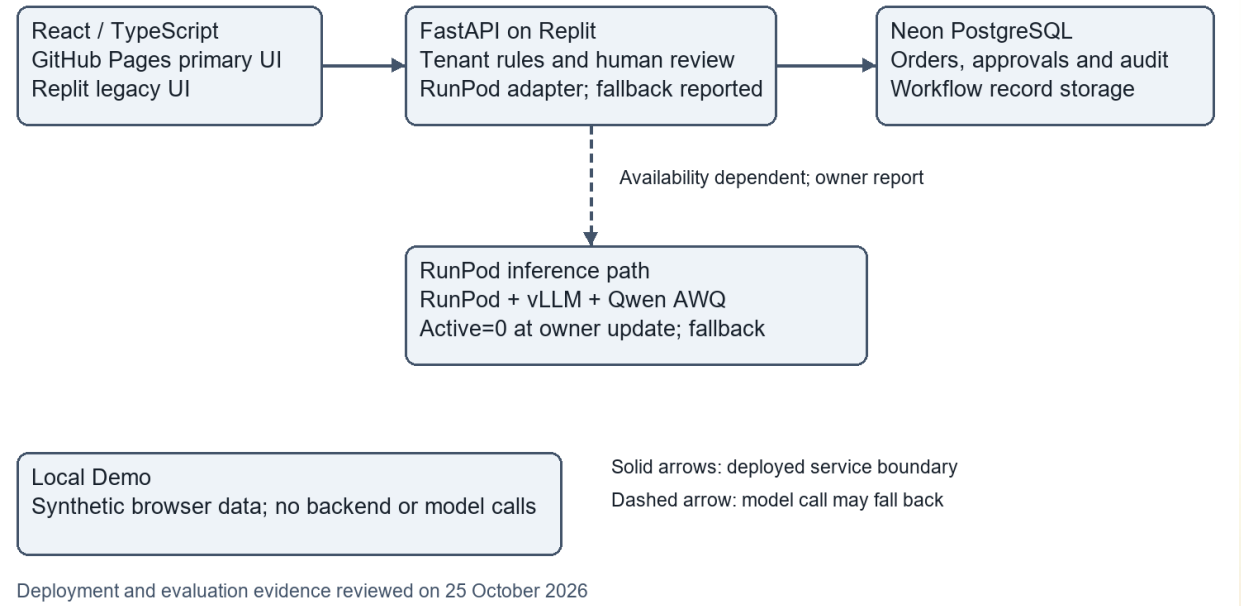


Figure A1. The database and AI service are separate backend branches. He Minghao reports a RunPod adapter configuration, Active=0 and public parse fallback. Standalone RunPod evaluation and browser-only Local Demo are distinct modes. Deployment status is owner-reported, not a new live test.

Evidence references

Official SDC Domain 2 Orbyte Smart SME Challenge brief, pages 3–4; Falcons planning information, 24 October 2026.

https://github.com/Hozier123/sdc-2026-team-project/pull/65 — approved and merged TEST32 freeze manifest; workflow execution remains separately reported.

https://github.com/Hozier123/sdc-2026-team-project/blob/main/docs/ops/stub-public-parse-gap.md — historical Stub configuration; superseded for deployment settings by He Minghao's supplied update of 25 October 2026.

https://github.com/Hozier123/sdc-2026-team-project/blob/main/evidence/EVIDENCE_INDEX.md — 23 live extraction cases; accuracy NOT_COMPUTED.

https://github.com/Hozier123/sdc-2026-team-project/blob/main/frontend/package-lock.json and backend/requirements.lock — dependency versions.